

Patient Record Pipeline

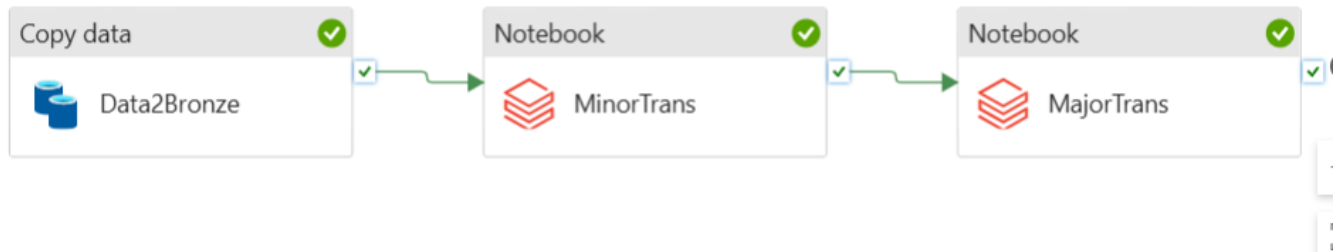
Screenshot Walkthrough

Azure Data Factory | Azure Databricks | Azure Key Vault | Power BI



Section 1: Azure Data Factory

Step 1 Pipeline Canvas — PatientRecordPipeline01



Step 2 Copy Activity — Source Tab (ds_pRecord_csv)

Connection	Schema	Parameters
Linked service *	ls_pRecord_csv Test connection Edit New Learn more	
File path	raw / Directory / Patient_Record.csv	
Compression type	No compression	
Column delimiter ①	Comma (,)	
Row delimiter ①	Default (\r,\n, or \r\n)	
Encoding ①	Default(UTF-8)	
Quote character ①	Double quote (")	
Escape character ①	Backslash (\)	
First row as header ①	☒	

Step 3 Copy Activity — Sink Tab (ds_pRecord_parq)



Connection

Schema

Parameters

Linked service *

Is_pRecord_parq

Test connection

Edit

New

Learn more

File path

patient-record

/

bronze

/

File name

Compression type

snappy

Step 4

Linked Services List

 Azure Data Lake Storage Gen2 [Learn more](#) 

ls_pRecord_csv

Description

Connect via integration runtime * ⓘ

 AutoResolveIntegrationRuntime 

Authentication type

Account key 

Account selection method ⓘ

☐ From Azure subscription ☒ Enter manually

URL *

https://retailsalesqb.dfs.core.windows.net/

Storage account key

Azure Key Vault

Storage account key *

.....

Test connection ⓘ

☒ To linked service ☐ To file path

Annotations

 New

 Azure Data Lake Storage Gen2 [Learn more](#) 

Name *

Is_pRecord_parq

Description

Connect via integration runtime * 

 AutoResolveIntegrationRuntime 

Authentication type

Account key 

Account selection method 

☐ From Azure subscription ☒ Enter manually

URL *

https://retailsalesgb.dfs.core.windows.net/

Storage account key

Azure Key Vault

Storage account key *

.....

Test connection 

☒ To linked service ☐ To file path

Annotations

 New

Step 5 Is_Minor Linked Service Detail

Name

Is_Minor

Description**Connect via integration runtime** * ⓘ

AutoResolveIntegrationRuntime

**Account selection method** *

From Azure subscription



Enter manually

Databrick Workspace URL * ⓘ

https://adb-7405613009689131.11.azuredatabricks.net

Authentication type *

Access Token



Access token

Azure Key Vault

AKV linked service * ⓘ

PatientKV

**Secret name** *

NewPatientKV



Edit

Secret version ⓘ

Latest version

**Step 6** Schedule Trigger — PatientData_Merge

Name *

PatientData_Merge

Description

Type *

ScheduleTrigger

Start date * ⓘ

5/5/2026, 2:30:00 AM

Time zone * ⓘ

Dublin, Edinburgh, Lisbon, London (UTC+1) ▾

ⓘ This time zone observes daylight savings. Trigger will auto-adjust for one hour difference.

Recurrence * ⓘ

Every 1 Day(s) ▾

▾ Advanced recurrence options

Execute at these times ⓘ

Hours

Minutes

Schedule execution times

02:30



Trigger Run Status

All	Schedule	Tumbling window	Storage events	Custom events	 Refresh
Local time : Last 24 hours Trigger name : All Status : All 					
Showing 1 - 1 items					
Trigger name 	Scheduled time	Trigger time 	Status 	Pipelines	
PatientData_Merge	5/7/2026, 2:28:00 AM	5/7/2026, 2:28:00 AM	 Succeeded	1	

Step 8 Pipeline Run History

Activity runs

Pipeline run ID 3c85403b-5250-46c7-a391-d863067ac27a

All status

[Monitor in](#)

Showing 1 - 3 of 3 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration run
MajorTrans	Succeeded	Notebook	5/6/2026, 2:00:57 AM	1m 12s	AutoResolveInt
MinorTrans	Succeeded	Notebook	5/6/2026, 2:00:04 AM	51s	AutoResolveInt
Data2Bronze	Succeeded	Copy data	5/6/2026, 1:59:41 AM	22s	AutoResolveInt

⚡ Section 2: Azure Databricks

Step 1 External Locations

Catalog Explorer >

External Data

External Locations ⓘ Credentials ⓘ Connections ⓘ

🔍 Filter locations

2 locations

Name	Credential	URL
 databricks_projects	databricks_projects	abfss://unity-catalog-storage@dbstoragezuopcatpryyvo.dfs.core
 patient_record_volume	volume_credential	abfss://patient-record@retailsalesgb.dfs.core.windows.net/bronz

Step 2 Cluster Configuration

Compute > Simple form: ON

sreelekshmi's Cluster 2026-04-30 21:54:56

ConfigurationNotebooks (0)LibrariesEvent logSpark UIDriver logsMetricsAppsSpark compute UI - Master

General

Compute name
sreelekshmi's Cluster 2026-04-30 21:54:56

Policy
Unrestricted

Performance

Machine learning

Databricks runtime
17.3 LTS (includes Apache Spark 4.0.0, Scala 2.13)

Preferred node type
Standard_D4ds_v416 GB Memory, 4 Cores

Terminate after

10

minutes of inactivity

Photon acceleration

Single node

Tags

No custom tags

> Automatically added tags

Step 3 Access Connector — Overview

unity_volume_connector

Access Connector for Azure Databricks

Search

RefreshDelete

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Automation

Help

Essentials

Resource group (move)
Projects

Location
UK South

Subscription (move)
Azure subscription 1

Subscription ID
61c349c4-4db7-492c-983c-31cdcce004fb

Tags (edit)
Add tags

State
Succeeded

Resource ID
/subscriptions/61c349c4-4db7-492c-983c-31cdcce004fb/r

Step 4 MinorT Notebook — Data Cleaning Code

Patient ID

Input Format - PT-19508, 00034824

Required Value - 19508, 34824

Remove duplicates

```
▶ ✓ Yesterday (2s) 15

df = df.withColumn(
    "patient_id_clean",
    F.regexp_replace(F.col("patient_id"), r"^(PT-|0+)", "").cast(IntegerType())
)

window_dedup = (
    Window.partitionBy("patient_id_clean")
    .orderBy(F.col("_file_modification_time").desc())
)

df = (
    df.withColumn("_row_rank", F.row_number().over(window_dedup))
    .filter(F.col("_row_rank") == 1)
    .drop("_row_rank")
)
```

Step 5 MinorT Notebook — Silver Table Verification

```
▶ ✓ Yesterday (1s) 44

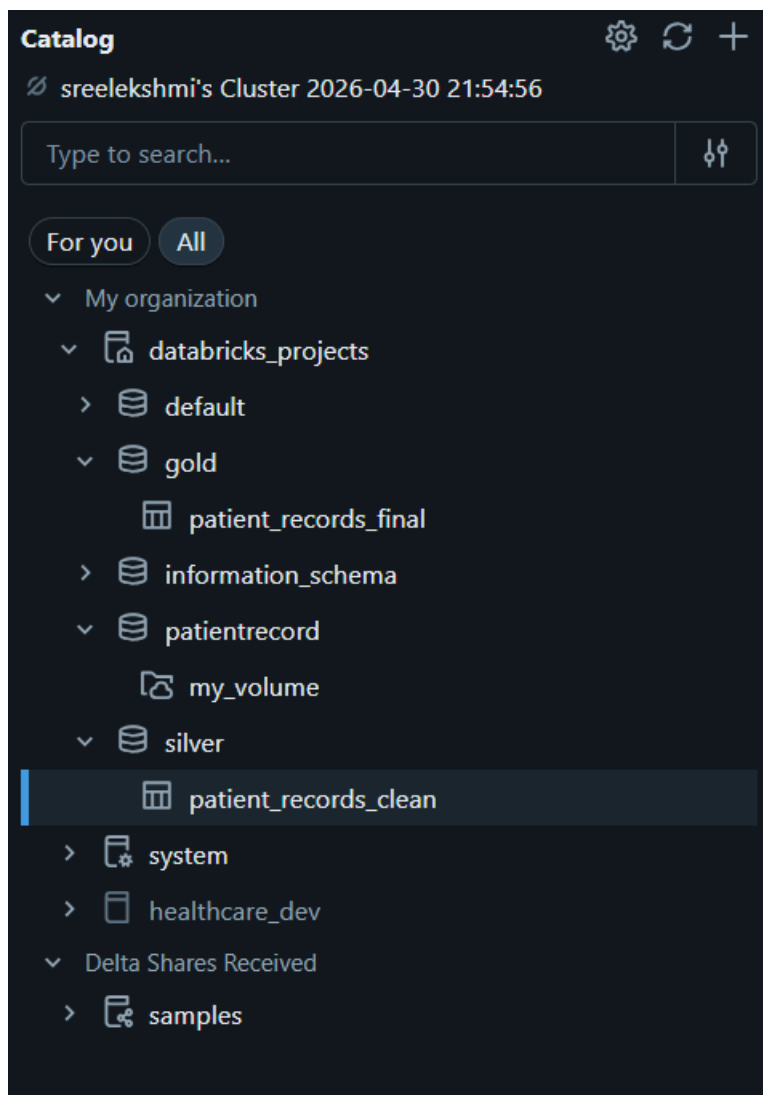
verify_df = spark.table(SILVER_TABLE)
print(f" Rows in {SILVER_TABLE}: {verify_df.count():,}")
display(verify_df.limit(5))

▶ (3) Spark Jobs
```

Step 6 MajorT Notebook — Delta MERGE Code

```
from delta.tables import DeltaTable
target_table = "gold.patient_records_final"
delta_table = DeltaTable.forName(spark, target_table)
(delta_table.alias("tgt")
 .merge(
     df.alias("src"),
     "tgt.patient_id = src.patient_id"
 )
 .whenMatchedUpdate(
     condition="tgt._silver_timestamp < src._silver_timestamp",
     set={
         "first_name": "src.first_name",
         "last_name": "src.last_name",
         "full_name": "src.full_name",
         "date_of_birth": "src.date_of_birth",
         "age": "src.age",
         "gender": "src.gender",
         "email": "src.email",
         "address": "src.address",
         "city": "src.city",
         "state": "src.state",
         "admission_date": "src.admission_date",
         "discharge_date": "src.discharge_date",
         "length_of_stay_days": "src.length_of_stay_days",
         "department": "src.department",
         "primary_diagnosis": "src.primary_diagnosis",
         "_silver_timestamp": F.current_timestamp()
     })
 .whenNotMatchedInsertAll()
 .execute())
```

Step 7 Unity Catalog — Silver and Gold Schemas



Step 8 Silver Table Schema — patient_records_clean

Catalog Explorer > databricks_projects > silver >

 **patient_records_clean**  

Use with BI tools

Share

Overview

Sample Data

Details

Permissions

Policies

History

Lineage

Insights

Quality

 AI Generating Description...

 Filter columns...



Column	Type	Comment	Tags	Column masking
patient_id	int			
first_name	string			
last_name	string			
full_name	string			
date_of_birth	date			
age	bigint			
gender	string			

Step 9

External Volume — my_volume

Catalog Explorer > databricks_projects > patientrecord >

my_volume

⋮

Share

Overview

Files

Details

Permissions

Description

AI generate

Add

/Volumes/databricks_projects/patientrecord/my_volume

Filter files and directories at this level

Name	Size	Last modified
Patient_Record.parquet	162.29 KB	32 minutes ago



Section 3: Azure Key Vault

Step 1 Key Vault Overview

Key vaults

...

How many Key Vaults do I have in total?

Default Directory onmicrosoft.com)

Create

Manage deleted vaults

Manage view

Refresh

Filter for any field...

Subscription equals all

Resource Group

☐	Name ↑	Type
☐	DataSecureProject1	... Key vault
☐	PatientRecordsKV	... Key vault

Step 2 Secrets List

PatientRecordsKV | Secrets

...

Key vault

<<

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Generate/Import

Refresh

Restore Backup

Manage delet

Name	Type	Status
AccessTokenPRecord		✓ Enabled
NewPatientKV		✓ Enabled


DataSecureProject1 | Secrets
☆ ...

Key vault

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Keys

Secrets

+

Generate/Import

↺

Refresh

↶

Restore Backup

🔑

Manage deleted secrets

Name	Type	Status
AzSQLPass		✓ Enabled
DBName		✓ Enabled
ServerName		✓ Enabled
UserName		✓ Enabled

Step 3 Linked Service

Azure Key Vault

Name *

PatientKV

Description

Azure key vault selection method ⓘ

☐ From Azure subscription ☒ Enter manually

Base URL *

https://PatientRecordsKV.vault.azure.net/

Authentication method

System-assigned managed identity

Managed identity name: **AzDataFactoryProject01**

Managed identity object ID: **3888055d-5551-4e17-9323-4bf49610a0e8**

Grant Data Factory service managed identity access to your Azure Key Vault. [Learn more](#) 

Test connection

☒ To linked service ☐ To secret

Annotations

 New

> Parameters

> Advanced ⓘ



Section 4: Power BI Report

Step 1 Report Dashboard View

